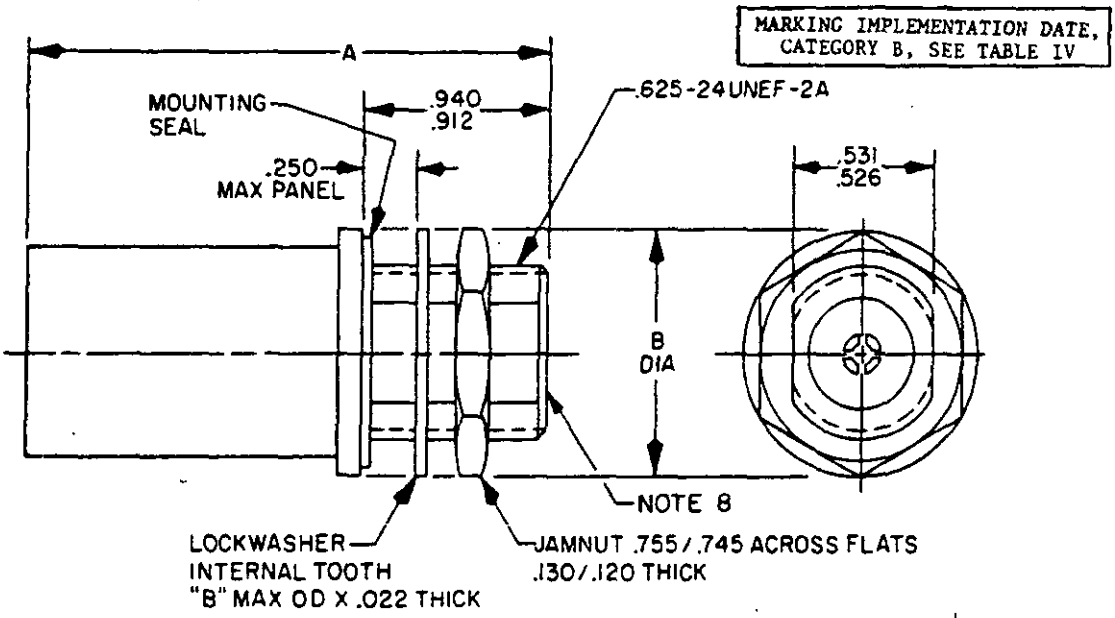
MILITARY SPECIFICATION SHEET

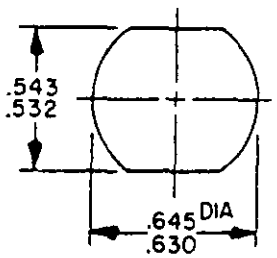
CONNECTORS, RECEPTACLES, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES N (CABLED), JAM NUT MOUNTED, SOCKET CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



Inches	mm	Inches	mm
.022	0.56	.543	13.79
.120	3.05	.625	15.88
.130	3.30	.630	16.00
.250	6.35	.645	16.38
.526	13.36	.745	18.92
.531	13.49	.755	19.18
.532	13.51	.912	23.16
		.940	23.88



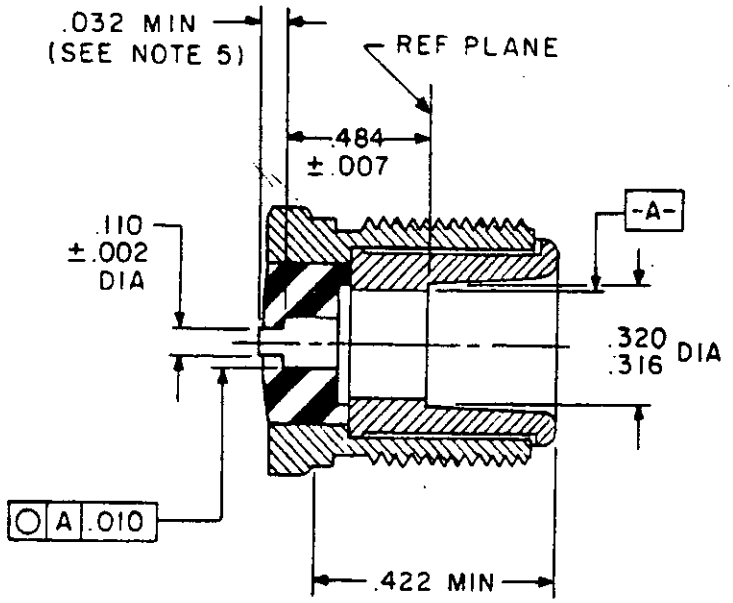
SUGGESTED MOUNTING HOLE

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See table I and III for dimensions A and B.
4. Dimensions A and B are the largest overall dimensions of the connector.
5. Wrench flats are to accommodate standard wrench opening in accordance with FED-STD-H28, appendix 10.
6. All undimensioned pictorial representations are for reference puposes only.
7. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
8. Series N, socket contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A
FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.



CATEGORY D

Inches	mm
.002	0.05
.007	0.18
.010	0.25
.032	0.81
.110	2.79
.316	8.03
.320	8.13
.422	10.72
.484	12.29

NOTES:
1. Dimensions are in inches.
2. Slot design optional.
3. Full threads to within .063 (1.58 mm) of shoulder; 1 1/2 maximum uneven threads to shoulder.
4. Metric equivalents are given for general information only.
5. Chamfer is optional, if chamfer is used put chamfer on a 30° maximum.

FIGURE 2. Category D captivation detail.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers, cross reference, and dimensions.

Dash number 1/	Applicable cable M17/#	Dimensions	Inches (millimeters) Minimum	Maximum
Category A - Field serviceable (no special tools required)				
0101 4/	2-RG6Δ 073-RG212* RG-222/U			
0012 4/	6-RG11Δ 62-RG144Δ 065-RG165 074-RG213 075-RG214* 77-RG216Δ	A B	.850(21.59)	1.900(48.26) .900(22.86)
0102 4/	078-RG217	A B	.850 (21.59)	2.094(53.19) .900(22.86)
Category C - Field replaceable (MIL-C-22520 crimp tool) 3/ See footnote next to applicable cable for crimp die				
0013 4/	073-RG212*≠ RG-222/U≠			
0014 4/	074-RG213*∢			
0015 4/	75-RG214*∢			
0018 4/	065-RG165*@∢	A B	.850(21.59)	2.031(51.59) .900(22.86)
0019 4/	RG-225/U*@∢			
0020 4/	6-RG11*Δ∢ 62-RG144*Δ@∢			
0021 4/	77-RG216Δ∢			
0023 4/	2-RG6*Δ≠			
Category D - Field replaceable - defined piece parts 3/5/2/				
0501 4/	75-RG214* RG-225/U@ 127-RG393	A B	.850(21.59)	1.875(47.62) .900(22.86)
0502 4/	074-RG213*			
0503 4/	060-RG142*@ 128-RG400	A B	.850(21.59)	1.781(45.24) .900(22.86)
0504 4/	028-RG058			

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

3/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

4/ These connectors have captivated center contacts.

5/ Not to be used in Army equipment.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and ∆.

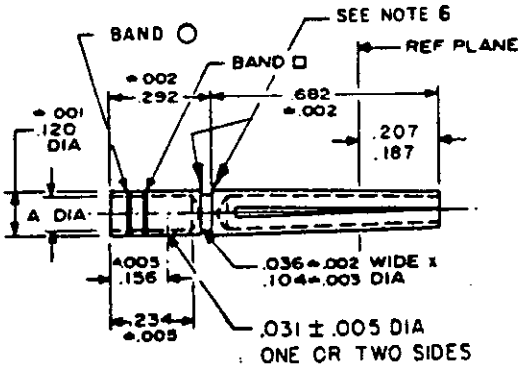
@ Cable to be used for the +200°C temperature cycling tests.

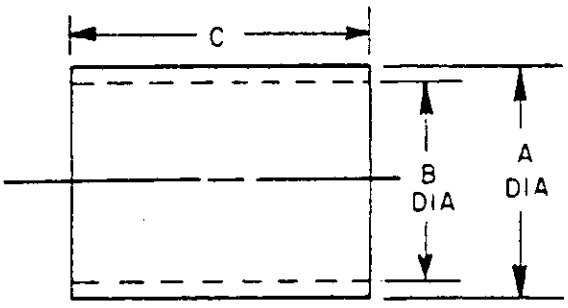
∆ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

≠ M22520/5-35 Closure A or M22520/5-55 Closure A

< M22520/5-61

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.





CRIMP FERRULE

Inches	mm
.001	0.03
.002	0.05
.003	0.08
.005	0.13
.015	0.38
.031	0.79
.036	0.91
.043	1.09
.098	2.49
.104	2.64
.120	3.05
.156	3.96
.187	4.75
.207	5.26
.220	5.59
.234	5.94
.250	6.35
.292	7.42
.418	10.62
.438	11.13
.492	12.50
.500	12.70
.600	15.24
.682	17.63

Dash No.	Ferrule No. 2/	A	B	C ±.015	Basic crimp tool 1/ M22520/5-	Crimp die or positioner M22520/5
0501	3-50	.492 +.003 -.000	.438 +.003 -.000	.600	-01	-25 Closure A or -61
0502	3-51		.418 +.003 -.000			
0503	3-52	.250 ±.003	.220 ±.003	.500		5, 11 or 57 Closure A or 19 Closure B
0504	3-53	.245 ±.001	.206 ±.001	.500		

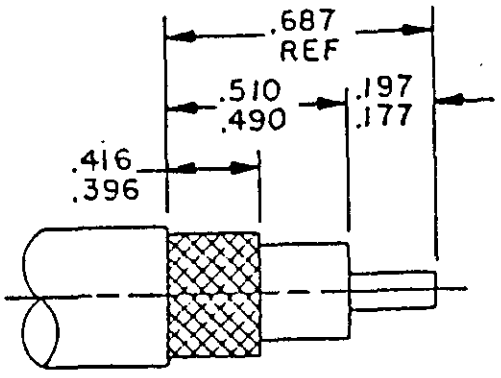
Dash No.	Contact No. 2/	A +.001 -.002	Basic crimp tool 1/ M22520/1	Crimp die or positioner M22520/1	Crimp tensile min	Color band □	Color band ○
0501 0502	3-10	.098 ±.002	-01	-14	60 lbs	Red	Red
0503 0504	3-11	.043 +.001 -.002	-01	-14	20 lbs	Blue	

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Center contact material shall be phosphor bronze or copper berylium.
4. Crimp tensile test shall be in accordance with MIL-C-39029.
5. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
6. .003 maximum break.

FIGURE 3. Contact and ferrule dimensions for category D only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.



Inches	mm
.177	4.50
.197	5.00
.396	10.06
.416	10.57
.490	12.45
.510	12.95
.687	17.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:
1,000 volts rms, maximum working voltage at sea level.

250 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 6 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque, 6 to 10 inch-pounds.

Mating characteristics:

Reference MIL-STD-348 and figure 2 for dimensions.

Center contact (socket):

Oversize test pin: 0.074 diameter, minimum (non-closed entry contacts only).

Insertion depth: .125, minimum.

Number of insertions: 1.

Insertion force test: Steel test pin diameter .066, minimum.

Test pin finish: 16 microinches.

Insertion force: 2 pounds, maximum.

Withdrawal force test: Steel test pin diameter .063, maximum.

Withdrawal force: 2 ounces, minimum.

Test pin finish: 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B, 5,000 megohms, minimum.

Center contact retention: (Applicable to captivated-center-contact connectors only). Fifteen pounds, minimum axial force for all cable except RG400 and RG142; 6 pounds, minimum for RG400 and RG142.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30, maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.008+.002 F (F in GHz).

Item 16: VSWR shall be less than 1.008+.002 F (F in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.012+.004 F (F in GHz).

Group B inspection: VSWR shall be less than 1.024+.007 F (F in GHz).

Qualification and group C inspection: VSWR shall not exceed 1.08.

Connector durability: 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum:

	Initial	After environment
Center contact	1.0	1.5
Outer contact	.2	Not applicable
Braid to body	.05	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202, 2,500 volts rms, minimum at sea level for connectors using other than RG400 and RG142; 1,500 volts rms, minimum for connectors using RG400 and RG142.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B . No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see tables I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 500 volts, minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,500 volts, rms tested at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

Cable retention force:

Noncrimp assemblies: 75 pounds, minimum.

Crimp assemblies:

50 pounds, minimum for cables .155 - .189 OD.

60 pounds, minimum for cables .190 - .229 OD.

75 pounds, minimum for cables .230 - .249 OD.

90 pounds, minimum for cables .250 OD and larger.

Coupling mechanism retention force: Not applicable.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss: .15 dB maximum tested at 10 GHz.

Part number: M39012/03 (dash number from table I or "B" number from table III).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

TABLE II. Group Qualification and retention testing. 1/

Group	Submission and qualification of any of the following connectors 2/	Qualifies the following connectors
	M39012/	M39012/
I	02-0003 02-0101 02-0104 03-0101 03-0102	02-0003 02-0101 02-0104 03-0101 03-0102
II	02-0003 02-0006 03-0012	02-0003 02-0006 03-0012
III	02B0007 02B0008 02B0009 02B0012 02B0013 02B0015 02B0016 02B0017 03B0003 03B0004 03B0005 03B0006 03B0008 03B0009	02B0007 02B0008 02B0009 02B0012 02B0013 02B0014 02B0015 02B0016 02B0017 02B0018 03B0003 03B0004 03B0005 03B0006 03B0008 03B0009 03B0010 03B0011
IV	02-0019 02-0020 02-0021 02-0022 02-0024 02-0025 02-0027 02-0028 03-0029 03-0013 03-0014 03-0015 03-0018 03-0019	02B0007 02B0008 02B0009 02B0012 02B0013 02B0014 02B0015 02B0016 02B0017 02B0018 02-0019 02-0020 02-0021 02-0022 02-0024 02-0025 02-0026 02-0027 02-0028 02-0029 02-0030 03B0003

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

TABLE II. Group qualification and retention testing 1/ - Continued.

Group	Submission and qualification of any of the following connectors 2/	Qualifies the following connectors
IV	M39012/	M39012/ 03B0004 03B0005 03B0008 03B0009 03B0010 03B0011 03-0013 03-0014 03-0015 03-0018 03-0019 03-0020 03-0021
V	02B0014 02B0018 03B0010 03B0011	02B0014 02B0018 03B0010 03B0011
VI	02-0026 02-0030 03-0020 03-0021 03-0023	02B0014 02B0018 02-0026 02-0030 03B0010 03B0011 03-0020 03-0021 03-0023
VII	03B0022	03B0022
VIII	03-0501 03-0502	03-0501 03-0502 03-0503
IX	03-0503	03-0503
X	03-0504	03-0504

1/ If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

2/ For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right hand column. The part does not necessarily have to be part initially qualified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

TABLE III. Category B - Nonfield replaceable (special tools may be required).

Not for Air Force or Navy use. For OEM use only.

Dash number 1/ 3/	Applicable cable M17/	Dimensions	Minimum	Maximum
M39012/03B0003 4/	073-RG212* RG-222/U			
M39012/03B0004 4/	074-RG213*			
M39012/03B0005 4/	75-RG214*			
M39012/03B0006 4/	078-RG217			
M39012/03B0008 4/	065-RG165	A B	.850(21.59)	2.031(51.59) .900(22.86)
M39012/03B0009 4/	RG-225/U*@			
M39012/03B0010 4/	6-RG11*Δ 62-RG144*Δ@			
M39012/03B0011 4/	77-RG216Δ			
M39012/03B0022 2/	2-RG6*Δ			

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ These connectors have captivated center contacts.

3/ For maintenance replacements for category B, see table V.

4/ Inactive for new design.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C temperature cycling tests.

Δ These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

TABLE IV. Cross reference of part numbers.

Preferred part number M39012/03	Substitute for part number or type designation 1/2/
-0101	UG-159D/U,M39012/03-0001
-0012	M23329/2-21,UG-160F/U, M39012/03-0002
B0003 3/	M23329/2-20, M39012/03-0003
	M23329/2-25
	M39012/03-0007
	UG-1699/U
	UG-1703/U
B0004 3/	UG-1700/U, M39012/03-0004
B0005 3/	M23329/2-22, UG-1701/U, M39012/03-0005
B0006 3/	M23329/2-23, UG-1702/U, M39012/03-0006
B0008 3/	M23329/2-26, UG-1704/U, M39012/03-0008
B0009 3/	M23329/2-27, UG-1705/U, M39012/03-0009
B0010 3/	M23329/2-28, UG-1706/U, M39012/03-0010
B0011 3/	UG-1819/U, M39012/03-0011
-0012	UG-1537A/U
-0013	M39012/03-0017
-0014	
-0015	
-0018	
-0019	
-0020	
-0021	
B0022 3/	M39012/03-0022
-0023	
-0501	
-0502	
-0503	
-0504	UG-556B/U

1/ The superseded part number or the type designation is for cross reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/02-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

3/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

TABLE V. Maintenance replacements for category B.

Category B number*	Category C dash number	Category A dash number	Category D dash number
B0003	0013	0101	---
B0004	0014	0012	0502
B0005	0015	0012	0501
B0006	---	---	---
B0008	0018	0012	---
B0009	0019	0012	0501
B0010	0020	0012	---
B0011	0021	0012	---
B0022	0023	0101	---

*Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3518-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:39Z
Check the source to verify that this is the current version before use.